Exp 3 MATLAB CODE

```
clc;
clear;
close all;

% Parameters
fs = 100000;            % Sampling Frequency
t = 0:1/fs:0.02;        % Time Vector
fm = 500;               % Message Frequency
fc = 5000;              % Carrier Frequency

% Message Signal
m = cos(2*pi*fm*t);

% Carrier Signal
c = cos(2*pi*fc*t);

% DSB-SC Modulation
dsb = m .* c;

% VSB Filter (Band-pass around carrier)
[b,a] = butter(4, [(fc-1000)/(fs/2) (fc+1000)/(fs/2)]);
vsb = filter(b,a,dsb);

% Coherent Demodulation
demod = vsb .* (2*cos(2*pi*fc*t));

% Recover Message (Low-pass filter)
```

```
[b1,a1] = butter(4, 1000/(fs/2));
rec = filter(b1,a1,demod);

% Display Results
figure;
subplot(5,1,1), plot(t,m), title('Message Signal'), xlabel('Time (s)'), ylabel('Amplitude'), grid on
subplot(5,1,2), plot(t,c), title('Carrier Signal'), xlabel('Time (s)'), ylabel('Amplitude'), grid on
subplot(5,1,3), plot(t,dsb), title('DSB-SC Signal'), xlabel('Time (s)'), ylabel('Amplitude'), grid on
subplot(5,1,4), plot(t,vsb), title('VSB Signal'), xlabel('Time (s)'), ylabel('Amplitude'), grid on
subplot(5,1,5), plot(t,rec,'r'), title('Recovered Message'), xlabel('Time (s)'), ylabel('Amplitude'), grid on
```

